

# Test Report

## Part I: Load Test

### 1. Test Back-end with npm load test

the target endpoint: <http://localhost/8080>

- 1) set **n = 10000**, **c = 1000** to evaluate the performance over huge number of requests and high concurrent requests scenario (this is very extreme test scenario)
- 2) According to the test result, [no errors reported](#), which indicates that the API is healthy and is handling all loads successfully and stably.
- 3) With a [mean latency of 2514.9ms](#), the API response time is relatively moderate, but if considering the extreme environment, its performance is pretty good.
- 4) [Requests per Second: 386](#) means the API can handle the loads well.
- 5) [the requests served within a certain time](#): suggest 99% of requests were processed in 2955ms or less.

### 2. Test Front-end with npm load test

the target endpoint: <http://localhost/3000>

- 1) set **n = 1000**, **c = 100** to evaluate the performance over more realistic number of requests and concurrent requests scenario
- 2) According to the test result, [no errors reported](#), which is excellent and provide users with good user experiences.
- 3) With a [mean latency of 141.9ms](#), which means each request took approximately 0.142 seconds to complete, that is health front end.
- 4) [Requests per Second: 386](#) means the API can handle the loads well.
- 5) [the requests served within a certain time](#): suggest 100% of requests were response to users in 185ms or less. This is totally positive result; users will obtain a good user experience.

## Part II: Functionality Test

- ✓ All users must **log in** before use the system. New use must sign up with his/her name and usask email address before log in;
- ✓ All users **can review, create, and select channel to review, create messages and replies.**
- ✓ All users can **post message/replies with or without screenshots.**
- ✓ The special user named **admin**, who has power to **remove users, channels, posts, and replies.**
- ✓ All users can **reply to replies** and all users can **view the visualized the nested replies.**
- ✓ All users can **approve or disapprove a post or reply** with a **thumbs-up/down with counter.**
- ✓ The same user cannot thumb up/down a same post/reply more than once.
- ✓ All users can **search** content by specific strings, the author and get the user with most/least posts, and the highest/lowest ranking of messages/replies.
- ✓ As an extension of functionality, the special user also can initialize database and deal with other users' ticket.